

HESHAM ABOUKEILA

POLYMER & MATERIALS ENGINEER

Houston, TX | +1 (405) 512-4519 | haboukeila@gmail.com | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Inventor on U.S. Patent No. 12,630,710 for enhanced polyester crystallization and developer of biobased films delivering 3x higher mechanical performance and 10x better barrier performance than commercial PBAT. Ph.D. chemical engineer specializing in polymer synthesis, processing, blown-film extrusion, structure-property relationships, surfactant formulation, and advanced characterization. Experience in materials innovation for sustainable packaging and circular economy solutions, spanning plastics upcycling, chemical recycling, and scalable product development.

TECHNICAL SKILLS

Polymer Processing: Extrusion/compounding, film blowing, compression molding

Materials Characterization: FTIR, DSC, TGA, GPC, SAXS, XRD, SEM, DMA

Performance Testing: Tensile testing, shear and extensional rheology, OTR/WVTR

Surface & Colloid: Surface/interfacial tension, contact angle, bubble pressure, foam analysis

Materials Expertise: Biobased and biodegradable polymers, flexible films, crystallization, structure-property analysis

Software: Aspen Plus, Aspen HYSYS, MATLAB, Symmetry, Python

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate | University of Oklahoma | Norman, OK

Jan 2025 - Jan 2026

- Characterized plastic-derived anionic surfactants synthesized from post-consumer recycled HDPE by measuring critical micelle concentration, surface and interfacial tension, foam stability, and calcium tolerance; demonstrated a CMC as low as 0.08 wt% and calcium tolerance of 221 ppm, competitive with conventional commercial surfactants, in a multi-institution plastics-upcycling collaboration.
- Led an industry-funded study with Church & Dwight benchmarking three commercial nonionic surfactants in potassium carbonate and sodium chloride electrolyte systems; expanded the usable HLD formulation range from approximately 4.5 to 8 and identified detergency optima of up to 90% soil removal, guiding chloride-free formulation strategies.

Graduate Research & Teaching Assistant | University of Oklahoma | Norman, OK

Aug 2021 - Dec 2024

- Engineered and evaluated 60+ biobased biodegradable polyester formulations as alternatives to LDPE/LLDPE for flexible-film packaging while onboarding and mentoring new team members and coordinating shared laboratory practices.
- Demonstrated 3x higher mechanical performance and 10x better barrier performance than commercial PBAT through thermomechanical, rheological, and permeability testing.
- Produced more than 20m of polyester and polyethylene films by lab-scale blown-film extrusion, resulting in an 80% reduction in permeability and 160% increase in tensile strength.
- Established a nucleating-agent strategy that accelerated crystallization by 56x, reducing crystallization half-time from 28 to 0.5 minutes.

Instructor | American University of the Middle East | Egaila, Kuwait

Sep 2019 - Aug 2021

- Delivered 7 engineering courses across 4 semesters, including 4 laboratory courses and 3 problem-solving courses, serving 150+ students per semester; used Aspen Plus and MATLAB to strengthen process-simulation and engineering problem-solving skills.
- Revived and independently delivered an additional course as course moderator and sole instructor, rebuilding the curriculum, lectures, assignments, and assessments from the ground up.
- Introduced an innovative remote-learning approach during COVID-19 by collaborating with faculty, laboratory staff, and the marketing team to produce 10+ laboratory-tour and equipment-introduction videos for Unit Operations and Fluid Mechanics.

Sales & Service Engineer | System Technique | Cairo, Egypt

Jul 2018 - Jul 2019

- Diagnosed and resolved system and process failures across water-treatment installations, including chemical systems, filters, pumps, and softeners, minimizing client downtime and maintaining service reliability.
- Tripled the active client base from 4 to 12 accounts within one year and met the annual sales target in 6 months by building customer-specific technical presentations and value propositions that translated technical performance into economic benefits.
- Spearheaded expansion beyond water-treatment chemicals and equipment into full-cycle swimming pool services, including design, construction, operation, and maintenance, opening a new revenue vertical for the business.

Graduate Research Assistant | NTNU / University of Bologna | Norway & Italy

Mar 2016 - Mar 2018

- Engineered Nafion/PEGDME hybrid membranes that increased dry-state CO₂ permeability by up to 36x compared with unmodified Nafion while achieving CO₂/N₂ selectivity above 30 under humid conditions.
- Controlled membrane nanostructure through solvent casting and water treatment, increasing CO₂ permeability by approximately 650% to 482 Barrer and CO₂/N₂ selectivity by 34%, surpassing the Robeson upper bound.

- Collaborated with international teams to quantify humidity effects on Aquivion membranes across six gases, demonstrating more than 10× higher permeability without sacrificing CO₂ separation selectivity.

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EDUCATION

Ph.D., Chemical Engineering University of Oklahoma Norman, OK	Dec 2024
M.Sc., Chemical Engineering University of Bologna Bologna, Italy	Mar 2018
B.Sc., Gas and Petrochemicals Engineering Alexandria University Alexandria, Egypt	Jul 2014

PATENT

1,5-Pentanediol-Based Polyester Compositions with Enhanced Crystallization Rates and Methods of Manufacture.
U.S. Patent No. 12,630,710 B1, 2026. [Patent link](#)

SELECTED PUBLICATIONS

- **Structure-Property Relationships in PLA-PBAT and PLA-PPeAT Blends: Toward Stronger Biobased Biodegradable Packaging Materials.** *Journal of Materials Science*, 2026. [DOI](#)
- **Closed-loop Recycling of Poly(pentylene adipate-co-terephthalate) via Amine-Catalyzed Methanolysis.** *ChemSusChem*, 2026. [DOI](#)
- **Enhanced Crystallization of Poly(pentylene adipate-co-terephthalate) via Ionomer Nucleation.** *Polymer Engineering & Science*, 2025. [DOI](#)
- **Film Blowing of Biobased Biodegradable Polyesters: Poly(pentylene adipate-co-terephthalate) and Poly(dodecylene furanoate).** *Macromolecular Materials and Engineering*, 2025. [DOI](#)
- **Biobased Poly(dodecylene furanoate) with Inherent Advantages in Performance and Circularity.** *ChemSusChem*, 2025. [DOI](#)
- **Upcycling of Waste Plastics into Carboxylic Acids for Biodegradable Surfactants.** *Angewandte Chemie*, 2025. [DOI](#)

Additional publications: [Google Scholar profile](#)

SELECTED HONORS

- Eastman Chemical Student Award in Applied Polymer Science Finalist, ACS PMSE Division (2024)
- Dissertation Excellence Award, Gallogly College of Engineering (2024)
- ANTEC Travel Award, Society of Plastics Engineers (2024)
- Judith K. Pearson Memorial Engineering Award (2023)

LEADERSHIP & PROFESSIONAL SERVICE

- President, Society of Plastics Engineers, University of Oklahoma Student Chapter (2022 - 2024)
- Chair, Academic Affairs Committee, Graduate Student Senate (2022 - 2024)
- Reviewer, Polymer Engineering & Science (2025 - Present)
- Reviewer, Journal of Polymer Engineering (2022 - Present)
- Officer, Chemical Engineering Graduate Student Society (2021 - 2024)
- Erasmus Exchange Student, Norwegian University of Science & Technology (2017 - 2018)
- Member, American Chemical Society and American Institute of Chemical Engineers

SELECTED PRESENTATIONS

- **Invited Talk, ACS Fall Meeting (2024):** Synthesis and Characterization of Biobased Copolyesters Based on Furandicarboxylic Acid: Poly(dodecamethylene furandicarboxylate)
- **Oral Presentation, AIChE Annual Meeting (2024):** Synthesis and Characterization of Biobased Copolyesters Based on Pentanediol Acid: Poly(pentylene adipate-co-terephthalate)
- **Poster, SPE ANTEC (2024):** Synthesis and Characterization of Biobased Copolyesters Based on Furandicarboxylic Acid: Poly(dodecamethylene furandicarboxylate)

LANGUAGES

Arabic (Native) | English (Fluent) | Italian (Beginner)